

蓝桥杯国赛JavaB组2023

2024-7-06 · marshall

本文记录了2023蓝桥杯国赛JavaB组的部分题解

1. 互质

```
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int MOD = (int)1e9+7;
    public static long qmi(int a, int k, int p) {
        long res = 1;
        while(k > 0) {
            if ((k & 1) == 1) res = res * a % p;
            a = (int)((long) a * a % p);
            k = k >> 1;
        }
        return res;
    }
    public static long gbd(long a, long b) {
        return b == 0 ? a : gbd(b, a % b);
    }
    public static void main(String[] args) {

        long res = qmi(2023, 2023, MOD);
        long a = qmi(2023, 2023, MOD) * qmi(7, MOD-2, MOD) % MOD;
        long b = qmi(2023, 2023, MOD) * qmi(17, MOD-2, MOD) % MOD;
        long c = qmi(2023, 2023, MOD) * qmi(17 * 7, MOD-2, MOD) % MOD;

        res = (res - a - b + c) % MOD;
        System.out.println(res);

    }
}
```

// 640720414

2. 逆元

```
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int MOD = (int)2146516019;
    public static long qmi(int a, int k, int p) {
        long res = 1;
```

```

        while(k > 0) {
            if ((k & 1) == 1) res = res * a % p;
            a = (int)((long) a * a % p);
            k = k >> 1;
        }
        return res;
    }
    public static long gbd(long a, long b) {
        return b == 0 ? a : gbd(b, a % b);
    }
    public static void main(String[] args) {
        long res = 0;
        for(int i = 1; i <= 233333333; i++) {
            res ^= qmi(i, MOD-2, MOD);
        }

        System.out.println(res);
    }
}
//

```

3.礼物

```

// 通过率20%
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int N = 100010;
    static int[] gifts = new int[2*N];
    public static void main(String[] args) {
        int n = sc.nextInt();
        for(int i = 0; i < 2 * n; i++) {
            gifts[i] = sc.nextInt();
        }
        Arrays.sort(gifts, 0, 2*n);
        long res = 0;
        for(int i = 0; i < n; i++) {
            int x = gifts[i] * gifts[2*n-i-1];
            res += x;
        }
        System.out.println(res);
    }
}

// 不开long long 见祖宗 , 通过率100%
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int N = 100010;
    static long[] gifts = new long[2*N];
    public static void main(String[] args) {
        int n = sc.nextInt();
        for(int i = 0; i < 2 * n; i++) {

```

```

        gifts[i] = sc.nextLong();
    }
    Arrays.sort(gifts,0,2*n);
    long res = 0;
    for(int i = 0; i < n; i++) {
        long x = gifts[i] * gifts[2*n-i-1];
        res += x;
    }
    System.out.println(res);
}
}
// 1307261675
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        System.out.println(1307261675);
    }
}

```

4.不完整算式

```

import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        char[] str = sc.next().toCharArray();
        String A = "", B = "", C = "", OP = "";
        int flag = 0;
        int j = 0;
        // 取A
        if (str[j] == '?') {
            flag = 1;
            j++;
        } else {
            while (Character.isDigit(str[j])) {
                A += str[j];
                j++;
            }
        }

        // 取OP
        if (str[j] == '?') {
            flag = 2;
            j++;
        } else {
            OP += str[j++];
        }

        // 取B
        if (str[j] == '?') {
            flag = 3;
            j++;
        } else {

```

```

        while(Character.isDigit(str[j])){
            B += str[j];
            j++;
        }
    }

    // 跳过=
    j++;

    //取C
    if (str[j] == '?') {
        flag = 4;
        j++;
    }else {
        while(j < str.length && Character.isDigit(str[j])){
            C += str[j];
            j++;
        }
    }

    if(flag == 1) {
        int b = Integer.parseInt(B);
        int c = Integer.parseInt(C);
        if (OP.equals("+")) {
            System.out.println(c - b);
        }else if (OP.equals("-")) {
            System.out.println(c+b);
        }else if (OP.equals("*")) {
            System.out.println(c/b);
        }else {
            System.out.println(c * b);
        }
    }else if(flag == 2) {
        int a = Integer.parseInt(A);
        int b = Integer.parseInt(B);
        int c = Integer.parseInt(C);
        if (a + b == c) System.out.println("+");
        else if (a - b == c) System.out.println("-");
        else if (a * b == c) System.out.println("*");
        else System.out.println("/");
    }else if (flag == 3) {
        int a = Integer.parseInt(A);
        int c = Integer.parseInt(C);
        if (OP.equals("+")) {
            System.out.println(c - a);
        }else if (OP.equals("-")) {
            System.out.println(a-c);
        }else if (OP.equals("*")) {
            System.out.println(c/a);
        }else {
            System.out.println(a/c);
        }
    }else {
        int a = Integer.parseInt(A);
        int b = Integer.parseInt(B);
        if (OP.equals("+")) {
            System.out.println(a + b);
        }else if (OP.equals("-")) {
            System.out.println(a - b);
        }else if (OP.equals("*")) {
            System.out.println(a * b);
        }
    }
}

```

```

        } else {
            System.out.println(a / b);
        }
    }
}
}

```

5. 星球

```

// 状态压缩DP
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int N = 20, n;
    static double[][] dp = new double[1 << N][N];
    static Pair[] stars = new Pair[N];
    public static void main(String[] args) {
        n = sc.nextInt();
        for(int i = 0; i < n; i++) {
            stars[i] = new Pair(sc.nextInt(), sc.nextInt(), sc.nextInt(), sc.nextInt());
        }
        for(int i = 0; i < 1 << N; i++) {
            Arrays.fill(dp[i], -1);
        }
        double res = Double.MAX_VALUE;
        for(int i = 0; i < n; i++) {
            res = Math.min(res, dfs(0, i));
        }

        System.out.printf("%.2f", res);
    }

    public static double dfs(int s, int i) {
        if (s == (1 << n) - 1) {
            return 0;
        }

        if (dp[s][i] != -1) {
            return dp[s][i];
        }

        double res = Double.MAX_VALUE;
        for(int j = 0; j < n; j++) {
            if ((s >> j) & 1 == 1) {
                continue;
            }
            res = Math.min(res, dfs(s | 1 << j, j) + stars[j].w * getDistance(stars[i], stars[j]));
        }

        return dp[s][i] = res;
    }
}

```

```

        public static double getDistance(Pair a, Pair b) {
            return Math.sqrt((a.x - b.x) * (a.x - b.x) +
                             (a.y - b.y) * (a.y - b.y) +
                             (a.z - b.z) * (a.z - b.z));
        }
    }

    class Pair {
        int x;
        int y;
        int z;
        int w;
        public Pair(int _x, int _y, int _z, int _w) {
            this.x = _x;
            this.y = _y;
            this.z = _z;
            this.w = _w;
        }
    }
}

```

6. 序列

```

// 勾八数论题
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int N = 100010, n;
    public static long gcd(long a, long b) {
        return b == 0 ? a : gcd(b, a % b);
    }

    public static long lcm(long a, long b) {
        return a / gcd(a, b) * b;
    }

    public static void main(String[] args) {
        long res = 0;
        n = sc.nextInt();
        for(int i = 1; i <= n; i++) {
            int x = sc.nextInt();
            res += (long)n * i / lcm(x, i);
        }

        System.out.println(res);
    }
}

```

7. 电动车

```

import java.util.*;

public class Main {

```

```

static final Scanner sc = new Scanner(System.in);
static int N = 200010,n,m;
static int[] p = new int[N];
static edge[] edges = new edge[N];
public static int find(int x) {
    if (p[x] != x) p[x] = find(p[x]);
    return p[x];
}
public static void main(String[] args) {
    n = sc.nextInt();
    m = sc.nextInt();
    int res = -0x3f3f3f3f,cnt=0;
    for(int i = 0; i < m; i++) {
        int a = sc.nextInt();
        int b = sc.nextInt();
        int w = sc.nextInt();
        edges[i] = new edge(a, b, w);
    }
    for(int i = 1; i <= n; i++) p[i] = i;
    Arrays.sort(edges,0,m);
    for(int i = 0; i < m; i++) {
        int a = edges[i].a;
        int b = edges[i].b;
        if (find(a) != find(b)) {
            p[find(a)] = find(b);
            cnt++;
            res = Math.max(res, edges[i].w);
        }
    }
    if(cnt < n - 1) System.out.println("-1");
    else System.out.println(res);
}
}

class edge implements Comparable<edge>{
    int a;
    int b;
    int w;
    public edge(int _a, int _b, int _w) {
        this.a = _a;
        this.b = _b;
        this.w = _w;
    }

    @Override
    public int compareTo(edge o) {
        return Integer.compare(this.w, o.w);
    }
}

```

8. 游戏

```

import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);

```

```

static int N = 100010,n,k;
static int hh = 0, tt = -1;
static int[] q = new int[N], a = new int[N];
static int[] t1 = new int[N], t2 = new int[N];
public static void main(String[] args) {
    n = sc.nextInt();
    k = sc.nextInt();
    for(int i = 0; i < n; i++) a[i] = sc.nextInt();
    long sumMax = 0;
    for(int i = 0; i < n; i++) {
        if (hh <= tt && q[hh] < i - k + 1) hh++;
        while(hh <= tt && a[q[tt]] <= a[i]) tt--;
        q[++tt] = i;
        if(i >= k - 1) sumMax += a[q[hh]];
    }

    hh = 0;tt = -1;
    long sumMin = 0;
    for(int i = 0; i < n; i++) {
        if (hh <= tt && q[hh] < i - k + 1) hh++;
        while(hh <= tt && a[q[tt]] >= a[i]) tt--;
        q[++tt] = i;
        if(i >= k - 1) sumMin += a[q[hh]];
    }
    // System.out.println(sumMax + " " + sumMin);
    double res = (sumMax - sumMin) * 1.0 / (n - k + 1);
    System.out.println(res);

}
}

```